Hearing

15a

Sound is produced by vibrating objects that cause the molecules in a transmitting medium (here taken as air) to vibrate. These vibrations are called sound waves. Sound waves travel through the air from a vibrating object to our ears.

Our ears can pick up different pitches. Pitches depend on the frequency of a sound wave – a high frequency vibration makes a high-pitched note, while a low frequency vibration makes a low-pitched note. Our ears can detect sounds ranging from 20 vibrations per second to 20 000 vibrations per second.

Below is a table detailing the different parts of a human ear.

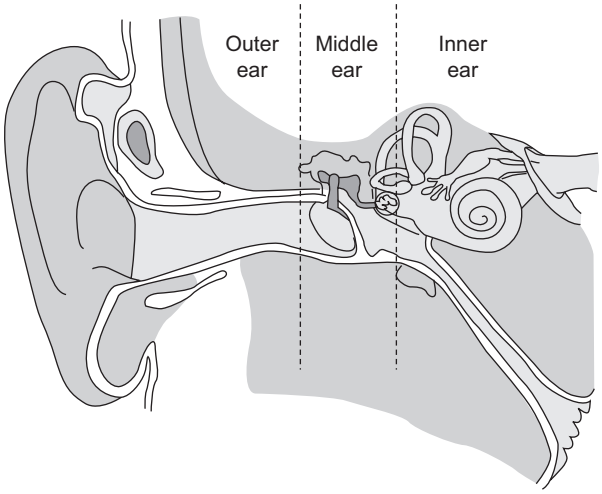
Part	Location	Function
Ear shell	Outer ear	Open funnel that collects sound waves
Ear canal	Outer ear	Long canal that leads sound waves to the ear drum
Eardrum	Middle ear	Thin membrane that vibrates when sound waves come in contact with it. It moves at the same frequency as the sound waves that hit it.
Hammer	Middle ear (surrounded by air)	Small bone near the eardrum. It picks up the vibrations from the eardrum and magnifies them.
Anvil	Middle ear (surrounded by air)	Hammer sends vibrations to the anvil, which is next to it. This is also a small bone that vibrates and magnifies vibrations.
Stirrup	Middle ear (surrounded by air)	Another small bone that picks up the vibrations from the anvil and magnifies them. It then vibrates against the inner ear.
Cochlea	Inner ear (filled with fluid)	Fluid filled tube that is coiled (looks like a snail shell). It contains receptors that detect vibrations in the fluid. Receptors send signals to the brain.
Semi-circular canals	Inner ear (filled with fluid)	Three tubes filled with fluid, each tube positioned in a different plane. They help with balance. Tiny hairs in the tubes trigger receptor cells that send signals about any movements to the brain.
Auditory nerve	Inner ear	Nerve that carries signals from the cochlea to the hearing centre of the brain. The brain then interprets these as sound.

1 **a** Use the table above. Which part of the ear has nothing to do with hearing? _____

b Explain why this part is very important. _____

_____

2 Use information from the table. Label the diagram of the ear.



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3 In your science book, rewrite the information given in the table as a report. Don't forget a title and an introduction. Use the lines below to plan your report.

4 When you have a middle ear infection, the middle ear partly fills with a thick liquid. Explain why hearing would be difficult for you if this happened.

5 If there is an extrememly loud noise near you, your ear may not operate properly.

a Describe what could have happened to your eardrum and explain what exactly would have caused that.

b Explain why there could be a ringing in your ears that starts in the cochlea.

6 A tube connects your middle ear with your mouth cavity. When you fly up in a plane or drive up a long, steep hill, your ears 'pop' as you open your mouth or swallow. Assume that air moves only slowly in that tube and explain what causes your ears to pop.

7 Somebody hits you on the head and you hear a ringing or buzzing noise. Your ears themselves are in good working order and there is no such noise outside. Explain where the noise may come from.

8 When you spin on the spot for a couple of minutes, the liquid inside the semi-circular canals spins with you. But when you suddenly stop spinning, the liquid doesn't, and you feel as if you are spinning the opposite way. Explain why you feel dizzy and find it difficult to walk properly.

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